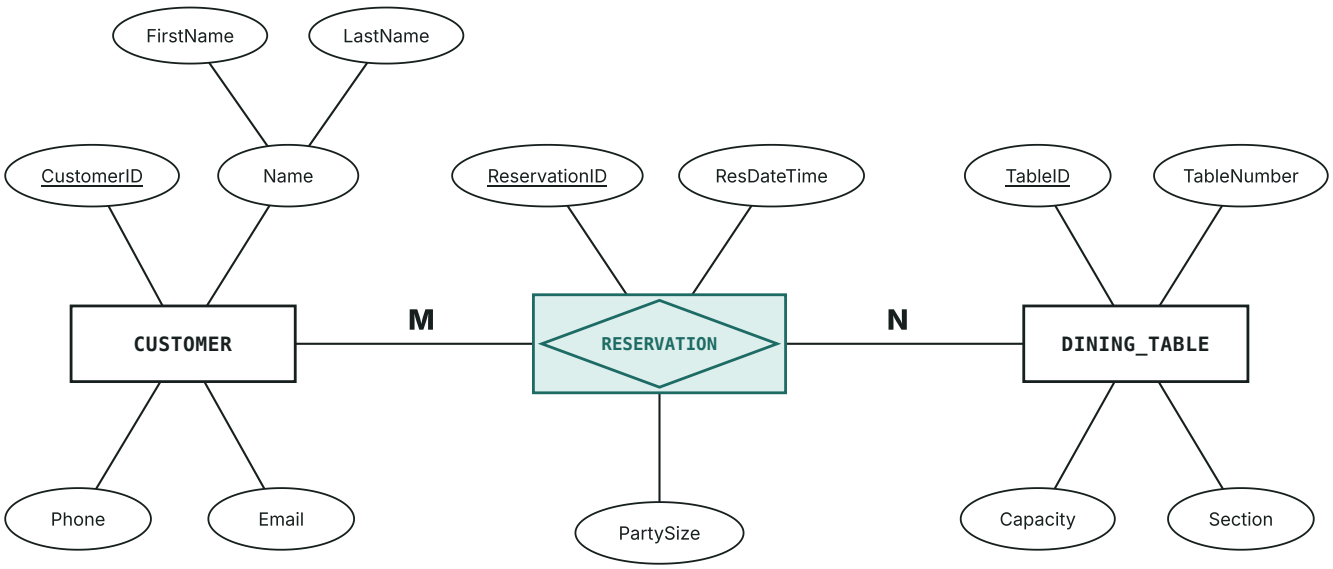
CSC 4500 · HW3 · ER MODEL DRAFT

Restaurant reservations in Chen notation

Customers reserve dining tables. A customer can book many tables over time, and a table is booked by many customers, so this is many-to-many (M:N). The reservation details (date/time, party size) belong to the relationship itself.



RESERVATION is drawn as an associative entity (a diamond inside a rectangle) because it is a relationship that also has its own key and attributes. If your professor prefers a plain diamond, label it `RESERVES` and hang ResDateTime and PartySize off it the same way.

How the diagram becomes MySQL tables

Each entity becomes a table. The M:N relationship becomes its own table that holds both foreign keys, which is your `ReservationTable`.

Customer	entity
CustomerID (PK)	INT AUTO_INCREMENT
FirstName	VARCHAR(40)
LastName	VARCHAR(40)
Phone	VARCHAR(15)
Email	VARCHAR(80)

ReservationTable	M:N bridge
ReservationID (PK)	INT AUTO_INCREMENT
CustomerID (FK)	→ Customer
TableID (FK)	→ DiningTable
ResDateTime	DATETIME
PartySize	INT

DiningTable	entity
TableID (PK)	INT AUTO_INCREMENT
TableNumber	INT
Capacity	INT
Section	VARCHAR(20)

Chen symbol key


Entity (a thing you store)


Relationship


Associative entity (relationship with its own data)


Attribute (a column)


Key attribute (underlined, becomes the PK)


Composite attribute (Name → First + Last)

Things to settle as a group

1. **Can a table be double-booked?** If not, add `UNIQUE (TableID, ResDateTime)` on ReservationTable.
2. **Can one reservation span several tables** (a party of 12 pushing tables together)? This diagram says no: one reservation, one table. A big party would get two rows.
3. **Total or partial participation.** Both legs are drawn as single lines (partial): a customer can exist with no bookings yet, and a new table can have none. Use a double line on a side if your rules say otherwise.
4. **Optional extra entity.** If you want more for queries, `SECTION` (Patio, Main, Bar) or `SERVER` could become its own entity linked 1:N to DINING_TABLE. Three tables already meet the assignment.